# Physicomimetics in 3D with real vehicles

The whole book in Bullet Physics Playground: the plan · 29 October 2026

This file maps every simulation, experiment and checkable result in *Physicomimetics: Physics-Based Swarm Intelligence* (Spears & Spears, Springer 2011) to a Bullet Physics Playground (BPP) port. It covers all 19 chapters and the appendix: 182 entries. Section 5 adds the ideas from the earlier chat, the plan and the thesis digests that go beyond the book. Section 6 extends the capstone mission to ground, surface and underwater vehicles, to show that the method is not tied to one vehicle. Section 7 proposes an escape from cul-de-sacs modelled on the chain fountain. The detailed notes for each chapter are in `book/` (`ch01.md` … `ch19.md`, `ch20-appendix.md`). They give the algorithm, parameter values, the results a port must reproduce (with page, figure and table references), and what the porter must decide where the book is silent or inconsistent.

Page numbers are PDF pages of `apbook1.pdf`. Book page = PDF page − 31.

**Status: done** = in BPP now; **partly** = in BPP but not yet checked against the book's numbers; **plan** = in `AP-3D-vehicles-plan.pdf`; **new** = not in the plan. A ★ marks an extension that is neither in the book nor in the app.

**Order inside each task:** port it first as a 2D particle replica and reproduce the book's numbers. Then fly it on vehicles. Then take it to 3D. A 2D replica is what makes a later vehicle or 3D result interpretable.

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## 1. What reading the whole book changed

These correct or extend the plan. Each is backed by the chapter notes.

1. **Ch. 13 is four mechanisms, not "friction".** In order of cost:
   - **AR damping η** (Eq. 13.9–13.10): scale the force by η = (|D − r|/D)^p_d for r < 2D, then stop the vehicle when |ΣF| falls below a threshold. This is a two-line change and the cheapest jitter fix.
   - **Adaptive friction** (Eq. 13.23), with k_more/k_less > 10.
   - **Static friction** (Eq. 13.25). Its printed condition is dimensionally odd; use μ·m·g < |ΣF·t̂|.
   - **Adaptive speed cap** near obstacles (Eq. 13.24). Its gain ratio is reversed: k_less/k_more > 10.

   The book gives ratios but no absolute gains. Build all four as separate switches (old behaviour at 0) and measure them one at a time. With η and the stop threshold the Pioneers "settle into fixed lattice formations" (PDF 454). Our expectation, not the book's, was that η alone makes the Newtonian drones settle like LJ, because both are then smooth through zero force at D. **Measured (M1, FINDINGS F43–F50): it does,** in 2D and 3D, as fast as LJ. With the book's band (r < 2D) a 2D launch settles at ~0.8 D; damping only within 0.2 D of D (William) keeps D. The stop rule adds nothing without η.
2. **Fixed-wing is not a dumbbell.** §13.5 says a vehicle that can only turn needs a *single* heading particle; that is the torque law of §13.3 (Eq. 13.15–13.18), which the chapter applies to its UAVs. The dumbbell (Eq. 13.19–13.22) is for vehicles that drive forwards and backwards.
3. **"APO from altitude" should build on Ch. 19, not Ch. 18.** Ch. 18's APO is an abstract 50-D optimiser: it collapses onto the best particle and shares fitness all-to-all. Ch. 19's APO is the triangle formation plus a pairwise fitness force (apoF, within 1.5D). It is local, deterministic and the same under rotation, and Ch. 19 calls it "a very simple modification" of the formation code. Sensing from altitude is our own idea (★); neither chapter mentions altitude.
4. **Kinetic theory (Ch. 4.4, Ch. 7) is velocity-based, with no forces.** Nothing in it oscillates about a switch, so it should not jitter on drones. That makes it a good contrast case for Ch. 13. Ch. 7 is entirely 2D, so 3D is our extension.

5. **The evolved chain law (Ch. 12) is nearly always repulsive.** This is our inference, not the book's statement: the evolved minor axis (95.7) exceeds the 60-pixel sensing range, so robots in range almost always repel, and the chain holds together through backflow pulses. Expect it to jitter on drones and to need the Ch. 13 mechanisms.
6. **Ch. 14 is not self-contained.** It does give the offline world (900×700, 40 robots, 90 obstacles, D 50, Vmax 20), the operators (fitness-proportional selection, one-point crossover, Gaussian mutation) and the fitness rule (three equally weighted penalties for collisions, cohesion and reaching within 4D of the goal; Eq. 14.5, PDF 477–480). It does not give the penalty sizes, population size, generations, mutation σ, the Daedalus reward update or a definition of "stuck". The only evolved LJ values in the book are on the Fig. 15.1 screenshot (PDF 519). Their robot–robot force cap (1.08) is about 3.6× below Fig. 14.3's (~3.9), while the obstacle cap (4.32) matches Fig. 14.4. Hettiarachchi's thesis fills some of this (see the thesis digest).
7. **BPP's `math.random` is the generator the appendix warns about; the shared generator replaces it (G2, done).** BPP links Lua 5.1 or 5.2, whose `math.random` calls glibc `rand()`, i.e. `random()`. Run inside Lua 5.1 with Listing A.1's sampling (S extra calls per accepted point, $10^6$ points, 36 bins), it fails at S = 25 ($\chi^2$ 321), 26 (223) and 27 (409) against a 99.9% cut of 66.6. The Marsaglia multiply-with-carry generator from `Uniform_Coverage.lua` passes at every S tried ($\chi^2$ 26–47). It now lives in `lib/rng.lua`, seeded from `SEED` or a `seed` parameter, and Uniform Coverage, the formations, `quadcopter/swarm.lua` and the probes all use it; `checks/rng_appendix.lua` is the regression test.
8. **The Maxelbot sensor is planar.** Its parabolic cones confine sound to a horizontal plane (Ch. 10.5.4.1). A Maxelbot-faithful 3D swarm must either fly in levels or assume a different (4-receiver) sensor. Treat that as a design decision, not an automatic upgrade.
9. **Clustering doesn't explain the 2D Newtonian jitter, but 3D needs a check.** Eq. 3.4 gives the clustering threshold G_t = Fmax·D^p/(2√3) for the triangle, and Eq. 3.5 gives Fmax·D^p/(2+2√2) for the square. The 2D drone defaults are below it (G 1500 against G_t 2368 triangle and 1699 square), so drones don't share nodes. The 3D square uses G 7500, which is above the 2D formula's 5492. A 3D G_t has not been derived, so measure it with a G sweep.
10. **Missing from the plan entirely:**
    - Ch. 2's integrator tests;
    - Ch. 4's Couette flow;
    - Ch. 5's local oriented potential fields (four formations without virtual colours, plus a gripper and a V that changes shape);
    - Ch. 10's sensor and vehicle stack;
    - Ch. 16's heterogeneous swarm design (rings, protection, surveillance, tracking);
    - Ch. 17's two perfect-hexagon methods;
    - Ch. 1's small demos (magnets chain, cooperative pull).
11. **The book's code listings contain bugs; don't copy them.**
    - Ch. 6: the APO sign in Listing 6.3 is reversed relative to Listing 6.4.
    - Ch. 7: speed `max(…, 6)` should be `min`, and Listing 7.3 draws cos θ = 1 − 2U, the 3D formula (its half-circle is made whole by its own assignment, F95).
    - Ch. 7: Kerr's theory numbers are his own error measures, not the book's, on a gas that misses the theory in three ways (F93).
    - Ch. 3: the Hooke cap applies to the attractive side only.
    - Ch. 13: the signs of Eq. 13.1 contradict the prose.
    - Ch. 18: Theorem 18.1's characteristic equation has a sign typo.
    - Ch. 9: the time units are inconsistent.

    Each chapter note lists its own.

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## 2. Shared groundwork

Most chapters need the same pieces. Build each once.

| ID | Piece | Needed by |
|---|---|---|
| G1 | **Verification monitors (done, `lib/monitors.lua`):** COM drift, linear and angular momentum, energy. A Lua integrator with controllable Δt for particle tests | 2, 3, 4, and every particle replica |
| G2 | **RNG module (done, `lib/rng.lua`):** move the multiply-with-carry generator from `Uniform_Coverage.lua` into a shared module, use it in the formations and the quadcopter harness, add a seed setting (not `os.time()`), and keep the appendix check as a regression test | 4, 7, 11, 12, 15, 17, 18, 19, A |
| G3 | **Headless batch runner (done, `lib/batch.py`):** seeds × configs, CSV out, 2 jobs in parallel, no drawing | 4, 6, 7, 9, 11, 12, 14, 15, 17, 18, 19 |
| G4 | **Velocity commands (done, `velocity()` in the vehicle interface):** fly this velocity, not this waypoint; the ideal vehicle, point mass, quad and boat; Couette flow on vehicles in `ch04/Couette-Vehicles.lua` | 4.4, 7, 11, 13, 17 |
| G5 | **Point-mass vehicle with limits (done, `lib/pointmass.lua`):** speed, acceleration and lag as parameters, for large swarms; the drone's by default (F87) | 7, 11, 15, 17 |
| G6 | **Scalar fields:** analytic functions, grids and diffusion, drawn as a ground texture or 3D slices | 1.5, 6, 8, 9, 10, 19 |
| G7 | **Neighbour sensing model:** range and bearing noise, maximum range, token-ring staleness (120 ms × N), 1-bit/packet payload, a Maxelbot preset calibrated to Fig. 10.14 | 3.4, 6, 10, 11, 12, 15 |
| G8 | **Rays:** line of sight between agents, and IR, sonar-ring and laser rangefinders (Bullet ray tests) | 5, 7, 9, 10, 11, 12, 13, 14 |
| G9 | **Force-law table** keyed by (source type, target type): asymmetric, with wildcards and virtual particles, and one sign convention | 3.4, 3.5, 5, 16, 17 |
| G10 | **Ground vehicles (Pioneer 3-AT done, `ground/`; Maxelbot to come):** skid-steer Maxelbot (MMP-5) and differential-drive Pioneer 3-AT | 9, 10, 11, 12, 13 |
| G11 | **Water vehicles (the ASV done, `boat/`):** a holonomic round ASV (Voith–Schneider); an ROV-style multi-thruster vehicle that can hover and move sideways; a torpedo-shaped AUV that cannot | 6, ★9, ★capstone |
| G12 | **Fixed-wing aircraft (kinematic model done, `fixedwing/`; Ch. 13 experiments in `ch13/`):** constant speed, bank limit, with the heading-particle torque law | 3.1 ★, 13 |
| G13 | **Worlds:** wall polygons from vertex lists, corridors, mazes, and an obstacle-density generator | 4.3, 7, 11, 12, 14, 15 |
| G14 | **Plume model:** Farrell-style puffs with wind, white-noise mixing and obstacles | 8, 9 |
| G15 | **Neighbour messages:** sensed values as well as positions; flooding broadcast | 4.1, 6, 9, 12, 19 |
| G16 | **Metrics (lattice quality started, `lib/lattice.lua`; drone settling in `probes/settle_drones.lua`):** lattice quality (angles, Delaunay CV), coverage grids and KL divergence, chain metrics (MST), suggested-versus-actual position gap, energy and path length, Q versus P | all |
| G17 | **Heading-dependent force terms:** elliptical zones, LOPF sectors | 4.2, 5, 12 |
| G18 | **One interface for every vehicle (done, `lib/swarm.lua`: the quadcopter, the ideal vehicle `lib/ideal.lua`, the ground robot and the boat; the laws shared in `lib/laws.lua`):** physicomimetics outputs a desired velocity; an adapter per vehicle turns it into commands. Vehicles that cannot move sideways use Ch. 13's heading particle or dumbbell | 13, ★capstone, every vehicle port |
| G19 | **Water:** the surface as a boundary, buoyancy, drag and currents (forces in Lua; BPP has no fluid simulation), and a plume carried by the current | 6, ★9, ★capstone |
| G20 | **Underwater sensing and comms:** sound only, ~1500 m/s, delayed and low-bandwidth; no GPS or radio below the surface; surface boats as gateways between sound and radio | ★6, ★capstone |

| ID | Piece | Needed by |
|---|---|---|
| G21 | **World generator (core and book presets done, `lib/world.lua`; realistic layers done: `lib/terrain.lua`, `lib/town.lua`, `lib/buildings.lua`, `lib/features.lua`, `lib/vegetation.lua`, with the five realistic presets in `lib/places.lua`; wind and currents later):** our own seeded, parameterised 3D generator, modelled on real places: terrain, water with depth, zones from rural to city centre, streets, lots, buildings of varied footprint and floor count, vegetation, and the capstone's harbour features. Presets: rural, suburban, city centre, industrial port, harbour town, and the book's worlds. Nothing from the CPT map makers; it writes Zarzhitsky's map format only so his plume tools can check our port (§6) | 3, 4, 7, 9, 11, 12, 14, 15, ★§7, ★capstone |

## 3. Overview by chapter

| Ch. | Title (author) | Entries | First BPP target | Vehicles | Groundwork | Status |
|---|---|---|---|---|---|---|
| 1 | Nature is lazy (Spears) | 8 | Potential-well height map; Q-vs-P metric | particles, ★ magnets | G6, G3 | new |
| 2 | NetLogo and physics (Spears) | 3 | Springs and solar system as integrator tests | particles | G1 | partly |
| 3 | NetLogo and physicomimetics (Spears) | 7 | Check the existing formations against the book; add Hooke, honeycomb, events | particles, quads | G1, G9, G16 | partly |
| 4 | Pushing the envelope (Spears) | 5 | Couette flow (2 closed forms); perfect square | particles, quads | G1, G3, G4, G15, G17 | partly / plan / new |
| 5 | Local oriented potential fields (Bravi et al.) | 9 | Square, line, star, tree by sector forces | holonomic particles, quads | G9, G17 | new |
| 6 | Bioluminescence (Karlsson et al.) | 7 | Slime-mould field + APO swarm, min and max tasks | particles, ASV, ★ AUV | G3, G6, G11, G15 | plan |
| 7 | Gas-mimetic swarms (Kerr) | 6 | Three theory checks, then KT vs Default vs Ant | particles, quads, ground robots | G3, G4, G8, G13 | partly |
| 8 | Fluxotaxis theory (Spears, Thayer, Zarzhitsky) | 6 | 3D closed-form unit tests | — | G6 | plan |
| 9 | Fluxotaxis experiments (Zarzhitsky) | 14 | Puff plume + 7-robot and swarm fluxotaxis | particles, Maxelbots, quads | G3, G8, G14, G15 | plan |
| 10 | What is a Maxelbot? (Maxim) | 13 | Trilateration noise model + token ring | Maxelbot | G7, G8, G10 | new |
| 11 | Uniform coverage (Maxim) | 13 | KL metric, wall-only control, f = 0.6 | particles, Maxelbot, quads | G3, G13, G16 | partly |
| 12 | Chain formations (Maxim) | 18 | Listing 12.4 maze with the evolved law | particles, Maxelbots, quads | G3, G7, G8, G13, G17 | plan |
| 13 | Motion control of constrained agents (Apker, Potter) | 13 | η, adaptive and static friction on the drones | quads, Pioneer, fixed-wing | G4, G10, G12 | partly (η on the drones; the Pioneer measured) |
| 14 | Adaptive learning, Daedalus (Hettiarachchi) | 7 | Offline three-law LJ obstacle course | particles, quads | G3, G8, G13 | plan |
| 15 | Statistical scouts (Spears et al.) | 10 | Bernoulli vs Bayes on the Ch. 14 world | particles (drones for show) | G2, G3, G13 | plan |

| Ch. | Title (author) | Entries | First BPP target | Vehicles | Groundwork | Status |
|---|---|---|---|---|---|---|
| 16 | Swarm design considerations (Wiegand, Ellis) | 9 | Type table; rings; protection | particles, quads | G9, G16 | new |
| 17 | Swarm engineering (Kazadi) | 9 | Hexagon Methods 1–2; quark model | particles, quads | G2, G3, G4 | plan / new |
| 18 | APO for global optimisation (Xie, Tan, Zeng) | 8 | Table 18.2 as a headless benchmark | — | G2, G3 | new |
| 19 | AP for noisy nonstationary environments (Spears) | 13 | apo.nlogo demo, N × D table, tracking | particles, quads | G3, G6 | plan |
| A | Anomalous random() (Spears, Green) | 4 | RNG regression test and demo | — | G2 | partly |

## 4. Every entry, chapter by chapter

Each row lists what the entry becomes in BPP. The chapter note has the full detail.

### Ch. 1 — Nature is lazy (PDF 34–57) · `book/ch01.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 1.1 | Swarm baseline P = 1 − Π(1 − $p_i$) versus measured Q | 37–38 | Standard summary line in the batch runner (G3) | new |
| 1.2 | Cooperative pull (video) | 38 | Ground robots or ★ quads with tethers moving a payload | new |
| 1.3 | Magnets self-organise into a chain (video; 39 + 1 dropped) | 42 | Rigid discs with dipole force and torque | new ★ |
| 1.4 | Compass construction of a triangular lattice | 44–45 | Ground-truth placement; ★ 3D sphere intersections | new |
| 1.5 | Rubber-sheet potential wells, incremental and batch | 45–48 | Live potential height map under 7 particles (G6) | new |
| 1.6 | MAV hexagonal sensing grid | 40 | Done as the drone triangle; add a grid-quality score | done |
| 1.7 | Least action | 41–44 | Energy and path length as standard metrics (G16) | new |

### Ch. 2 — NetLogo and physics (PDF 58–85) · `book/ch02.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 2.1 | `spring1D` : Hooke oscillation, momentum 0, energy flat, friction drains it | 58–72 | `ch02/` : scene and checks, all results reproduced | done |
| 2.2 | `spring2D` : angular momentum conserved, centrifugal stretch > 2D | 73–77 | `ch02/` : scene and checks, reproduced; ★ 3D off-plane spin; ★ spinning tetrahedron (tennis-racket flips); ★ 3D spring lattice | done |
| 2.3 | `solar_system` : Δt failures, velocity cap fails, force cap works (precession) | 78–85 | `ch02/` : scene and checks, all five experiments reproduced (peak speeds 5.5, 5.0, 4.0, 9.0 as printed); ★ several planets on tilted elliptical orbits, stable with a gap of 1.8 between orbits and ~50 steps per nearest-point pass (F58) | done |

## Ch. 3 — NetLogo and physicomimetics (PDF 86–123) · `book/ch03.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 3.1 | Boids ( `flock` ): turn caps 5/3/1.5°; snapshots at 546, 1,309, 78, 42 steps | 86–95 | Constant-speed particles; ★ 3D boids on quads or fixed-wing | plan |
| 3.2 | Hooke lattice ( `formation_hooke` ): k 0.5, Fmax 1; fails at Fmax 3 and 5; kill and birth | 96–103 | `ch03/models.lua` : the lattice by 200 steps and every failure reproduced ( `checks/ch03.lua` , F75); drones to come | partly |
| 3.3 | Split-Newtonian triangle, G_t = Fmax·D^p/(2√3), hysteresis, 50 → 21 nodes | 103–108 | Done; G_t, hysteresis and robustness confirmed on the model and the BPP scene; 3D G_t = G_t/√2 derived and measured ( `ch03/` , F73) | done |
| 3.4 | Square with two types, defect removal by noise, 500 particles | 108–113 | Done; Eq. 3.5 fits the Java/BPP force direction, not the .nlogo's (F74); add annealing, type flips, toggling, a quality metric; ★ NaCl in 3D | partly |
| 3.5 | Honeycomb: √3 renormalisation, 120°/60° repair, square↔honeycomb toggle | 113–115 | `ch03/models.lua` ( `honeycomb.nlogo` line for line): centres trap particles, the model's switching (a random mix of the two rules per pair, F77) frees them; McLurkin's repair not given in the text; drones at one altitude, ★ diamond lattice to come | partly |
| 3.6 | Goal and obstacles; Disable One; abandon threshold goalF > Fmax | 115–121 | Done; Figs. 3.24–3.26 reproduced on the model; the threshold is exact for a lone neighbour, a bound in a formation (F75) | done |
| 3.7 | Hardware requirements (GPS, IR, trilateration) | 121–123 | Sensing model (G7, G8) | new |

## Ch. 4 — Pushing the envelope (PDF 124–157) · `book/ch04.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 4.1 | Perfect formations ( `formation_perfect` ): (m, n) slots; the triangle breaks Newton's third law | 124–130 | `ch04/perfect.lua` (the `.nlogo` line for line): both lattices perfect, the momentum signature, One is Born, live D (F82); toggles rebuild turned 11–15°; question (a): the third law can be kept, 5–12× slower (F83); ★ (m, n, k) cube; questions (b), (c) open | done |
| 4.2 | Chains ( `formation_chain` ): elliptical law; Table 4.1 (561 → 350) | 131–138 | Particles with heading (G17), 50-run batches; ★ prolate spheroid in 3D | plan |
| 4.3a | Coverage baseline ( `uniform` ): 84×84, 9 cells; Table 4.2 | 139–144 | `ch04/coverage.lua` ( `uniform.nlogo` line for line): Table 4.2 and Fig. 4.17 reproduced (F78) | done |
| 4.3b | Mean free path ( `uniform_mfpl` ): bound 17⅓; Fig. 4.19 dip at 53–65 | 144–147 | Fig. 4.19 and its dip reproduced; William's NetLogo runs reproduced at MFP ≥ 17, below it noise-limited at 2 × $10^8$ agent steps; the BPP scene matches his Android runs ( `ch04/` , F78); one agent run 5 × $10^9$ steps: below 0.003 for MFP ≤ 28, the bound confirmed (F84); ★ 27 cells in 3D | done |
| 4.4 | Couette flow ( `kt` ): slope V/6 with r > 0.999; ν̄ = 1.2617√T (theory 1.2533) | 147–156 | `ch04/couette.lua` : profile and momenta reproduced; the `.nlogo` 's walls run hot (slope +1–2.5%, ν̄ 1.36√T); inward-only walls (Kerr's, the default) give Fig. 4.23 (F85, F86); on point masses and quads with velocity commands (F88–F91); ★ 3D slab | done |

## Ch. 5 — Local oriented potential fields (PDF 160–175) · `book/ch05.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 5.1 | Square deployment by 8 sector forces (no virtual colours) | 164–165 | Particles → drones; compare defects with Ch. 3's square | new |
| 5.2 | Linear deployment | 165 | Same machinery | new |
| 5.3 | Star with leader, arms and levels | 166 | Same; ★ 3D star with 4 or 6 arms | new |
| 5.4 | Star "gripper" (arms closing by ±γ) | 171 | ★ Drones grasping a rigid payload | new |
| 5.5 | Tree / V with levels | 167 | ★ Flying V of drones | new |
| 5.6 | V that changes shape (α changed at run time) | 171 | Same | new |

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 5.7 | Matarić behaviours in AP (wander, follow, disperse, aggregate, home, angularity) | 162–168 | A behaviour library on the formation code | new |
| 5.8 | Holes and choice of starting formation | 171–173 | Measure holes against the starting lattice | new |
| 5.9 | "Angle of attention" (occluded view) | 174 | Ray-cast occlusion share per formation (G8) | new |

The chapter gives no numbers, so the port defines its own parameters and metrics.

## Ch. 6 — Bioluminescence (PDF 176–223) · `book/ch06.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 6.1 | Dinoflagellate field generator (slime mould, 290/400/510 dinos) | 183–188, 200 | Grid field + drawing (G6); ★ 3D volume, currents | plan |
| 6.2 | ASV swarm: split-Newtonian + APO + goal (only drones with neighbours feel the goal) | 188–192 | Particles → ASVs | plan |
| 6.3 | Force balance: goalF < 0.2598, apoF < 0.13 | 196–198 | Batch sweeps | new |
| 6.4 | Max/min tasks: rising H1, U-shaped H2 with N = 6 best; time curves; 1295/1313/1292 ticks with no chemical | 198–209 | ~12,000 particle runs (G3); then ASVs | plan |
| 6.5 | Multi-drone simulator (asynchronous clocks, comms, fuel) | 210–218 | Our harness plus G7, G15 and fuel accounting | plan |
| 6.6 | 20-drone plume demo | 216–217 | Demo | plan |
| 6.7 | Holonomic ASV hull (18″, two Voith–Schneider propellers) | 218–222 | `boat/vehicle.lua` (G11): the book's hull, mass and thrust chosen; follows AP like the drones (F67); ★ AUV next | done |

## Ch. 7 — Gas-mimetic swarms (PDF 224–253) · `book/ch07.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 7.1 | KT controller: Rayleigh/Gaussian wall rebound, 15″ collision zone, 24-sonar ring | 228–236 | `ch07/kerr.lua` : Kerr's 2004 gas simulator (Garcia's DSMC in 2D; his Java in `ch07/kerr/` ), ported exactly, his departures as switches (F92, F93); the robot controller (his 2005 `KTRobot` ) next, then velocity-commanded quads (G4, G8) | partly |
| 7.2 | Uniform density at rest, error < 10% | 239–240 | `ch07/` : his box piles the gas into a corner (36% by the book's measure, 0.19 by his); with Garcia's wall at y, uniform to 2% at N = 10,000, 5–6% from N = 500 (F93); Fig. 7.5 is a later simulator's (F94) | done |
| 7.3 | Mean speed versus T, error < 2% | 240–241 | `ch07/` : his gas runs 3.75% hot, ~0.8% by his measure (Fig. 7.6's level); with three fixes 0.3% (F93); ★ 3D Maxwell √(8kT/πm) | done |
| 7.4 | Profiles with one moving wall (linear) and two (flat), against obstacle density | 241–243 | `ch07/` : both, explicitly, without obstacles: two-sided flat at V, one-sided linear with slip, error 20% → 4% from Mach 1 to 8 settled (his window 51% → 6%; F92, F93); Figs. 7.7–7.8's points read from their source; obstacles not in his code, open (F94) | partly |
| 7.5 | Default and Ant baselines | 244–246 | Particles; pheromone grid | new |
| 7.6 | KT vs Default vs Ant: KT covers ~90% in about a third of Ant's time | 246–251 | 1000″×5000″ corridor, 50×250 grid; ★ 3D canyon or tunnel | plan |

## Ch. 8 — Fluxotaxis theory (PDF 254–281) · `book/ch08.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 8.1 | Task split; the swarm as a finite-difference grid | 256–264 | Design note | plan |

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 8.2 | Emitter test: positive outward flux for time τ | 267–268 | Termination rule for Ch. 9 | new |
| 8.3 | GDMF navigation law | 269–271 | Core law | plan |
| 8.4 | Anemotaxis overshoots; fluxotaxis stops at the emitter | 272–273 | Analytic test | new |
| 8.5 | Continuous emitter: GDMF ∝ $r^{-3}$ against gradient ∝ $r^{-2}$ | 275–277 | 3D closed-form unit test | new |
| 8.6 | Single puff: gradient → 0 at the emitter, GDMF → ∞ | 277–280 | 3D unit test (the note re-derives the garbled formula) | new |

## Ch. 9 — Fluxotaxis experiments (PDF 282–331) · `book/ch09.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 9.1 | 1 versus 3 Maxelbots on ethanol: 22% vs 60% success | 285–286 | Small hardware replica (G10, G14) | new |
| 9.2–9.3 | Smoke around a robot; 32-sensor flume | 287–288 | ★ downwash stirring the plume | new |
| 9.4 | Finite-difference stencils | 291 | Library | plan |
| 9.5 | Puff plume model | 291–295 | G14, the largest single build. Zarzhitsky's CPT code (PlumeGen, TraceGen; kept outside the repo) gives the constants the book omits; the Android app's Chemical Plume Tracing is ported from it with a changed generator. Run the original on our generator's maps as an independent check | plan |
| 9.6 | LJ variant F = D/r² − D^1.7/r^2.7 | 296–299 | Add to the force-law table | new |
| 9.7 | Simulator loop, collision avoidance overrides tracing | 297–300 | Harness | plan |
| 9.8 | 7-robot hexagon: casting, chemotaxis, anemotaxis, fluxotaxis | 300–306 | Particles → quads | plan |
| 9.9 | Swarm versions (45° casting, flux toward each neighbour) | 306–312 | Unchanged in 3D: the cheapest route to 3D | plan |
| 9.10 | Exp. 1: success 74.6% fluxotaxis (Table 9.3) | 312–315 | Batch | plan |
| 9.11 | Exp. 2: obstacles, emitter enclosed 49.1% of the time | 315–318 | Batch | plan |
| 9.12 | Exp. 3: 7–70 robots, 40,500 runs (Tables 9.4–9.5) | 318–323 | Batch (subset) | plan |
| 9.13 | Exp. 4: 10–150 robots (Table 9.6) | 323–326 | Batch; ranking reverses against 9.12 (book inconsistency) | plan |
| 9.14 | Urban demo | 328–329 | ★ 3D city with source height | plan |

## Ch. 10 — What is a Maxelbot? (PDF 332–371) · `book/ch10.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 10.1 | Closed-form trilateration (6″ L) | 333–335 | Sensor model core; ★ 4-receiver 3D version | new |
| 10.2 | Multi-hop coordinate sharing | 335 | G15 | new |

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 10.3–10.5 | SRF04 accuracy; wavefront jitter fixes; range 2 → 4 → 9–11 ft | 336–344 | Noise and range parameters | new |
| 10.6 | Parabolic cones: sensing is planar | 345–346 | Elevation limit in 3D (decision) | new |
| 10.7–10.8 | RF Manchester link; module timing | 345–351 | Latency and drop model | new |
| 10.9 | Grid test: mean error 0.6″, max 1.2″; ≤ 11% at 9–11 ft | 351–352 | Calibration target for G7 | new |
| 10.10 | Token ring, 120 ms per slot, recovery | 352–355 | Staleness scheduler; ★ how large N can grow at drone speeds | new |
| 10.11 | One robot moves at a time (centralised design) | 356–357 | ★ Sequential update mode as a jitter test | new |
| 10.12 | Localisation I/O (x, y, heading byte) | 357–361 | Neighbour-table format | new |
| 10.13 | MMP-5 skid-steer base, IR 10–80 cm, compass, temperature; 26″ follower | 361–370 | Ground vehicle (G10); ★ heat-source follower | new |

## Ch. 11 — Uniform coverage (PDF 372–397) · `book/ch11.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 11.1–11.3 | Maxim's simulator, cell tiling, KL metric (natural log) | 373–376 | Metric (G16) | partly |
| 11.4 | Markov-chain existence proof | 376–377 | Analytic test | new |
| 11.5 | Turns at walls only: centre least visited (negative control) | 378–379 | Particles | new |
| 11.6 | Mean-free-path turns at f = 0.6 × cell: KL 0.000060 | 379–381 | Particles (ported, check KL); Maxim's simulator: 3×3 at f = 0.6 KL 0.00008, f from 0.05 to 4 on all 12 environments, 0.65 an upper bound ( `ch11/` , F84) | partly |
| 11.7 | Mean free path theory 0.6518 (0.6079 with buffer) | 380–383 | `ch11/` : 2D reproduced (exact 0.651757, buffer 0.6079); ★ 3D 0.7720 × edge (buffer 0.7138), with a closed form | done |
| 11.8 | 12 environments × 5 cell sizes (Table 11.1) | 383–386 | `ch11/` : Maxim's simulator and environments (William, 2026-10-25) and an exact Lua copy; every entry from his files, one misprint (F79, F80); rerun 40–160M steps: the small environments settle, the large ones' KL keeps falling with run length; his robot can stick on a wall (F81) | done |
| 11.9 | Multi-robot formula 1 − (1 − 1/C)^N | 384 | Check with N drones ★ | new |
| 11.10a–c | Hardware runs: 5×5 ft, 5×15 ft, L-shape | 384–396 | Maxelbot replica (G7, G10) | new |
| — | 3D coverage, altitude bands, per-minute scoring | — | ★ from the plan | plan |

## Ch. 12 — Chain formations (PDF 398–443) · `book/ch12.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 12.1–12.3 | Elliptical law, backflow force, front tangential force | 399–402 | Particles (G17) | plan |
| 12.4 | EA with 11 evolved parameters (Listing 12.2) | 402–404 | Use the published values | plan |

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 12.5–12.7 | Maxelbot simulator, 35-vertex maze (Listing 12.4), metrics, 9 environments | 404–413 | G13, G16 | plan |
| 12.8–12.10 | Exp. I scalability, II noise, III add then kill | 414–418 | Batch, 10 seeds | plan |
| 12.11 | 35-step backflow pulse test | 419 | Check | new |
| 12.12 | Ideal chain length (closed form) | 419–421 | Check (the book's example gives 37; the formula gives 38.5) | new |
| 12.13–12.16 | Chain-list protocol, display, laptop link, token loss | 422–428 | G15; ★ chain as a comms relay | new |
| 12.17 | Static Maxelbots: line, L, U at 2, 4, 6 ft | 429–436 | Replica (G10) | new |
| 12.18 | Dynamic: open floor, 5×14 ft corridor, L corridor | 436–442 | Replica; then ★ 3D shafts on quads | plan |
| ★ | Chain-fountain escape (§7): the same idea of a chain, used to pull agents out of a trap rather than to deploy a relay | — | Compare with the Ch. 12 law | new |
| ★ | Chain applications: relays in tunnels, mines, caves, garages, urban canyons, stairwells, over buildings, into pits; maze exploration; map carried back to base; branching at junctions; loops that reconnect; wall maps from range hits. Score: does the line-of-sight link hold? | — | Quads after the 2D replica (see §5) | new |

## Ch. 13 — Motion control of constrained agents (PDF 444–471) · `book/ch13.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 13.1–13.2 | Point-mass reachable space; the discontinuity at D | 447–448 | Explains the Newtonian jitter | done (observed) |
| 13.3–13.4 | Differential-drive and Pioneer reachable space | 449–453 | `ground/vehicle.lua` (G10): the Pioneer with the book's r_diff, wheel range and caps; read as stopping the wheel, the minimum speed freezes the lattice as Fig. 13.7 predicts (F66); read as turning it at 0.1 m/s, the robots reach D and get through the obstacle field (F72); plots to come | partly |
| 13.5 | **AR damping η + stop threshold** | 453–454 | `lib/ch13.lua`; on the drones `etaPower` 1, `etaWidth` 0.2 (2D) / 0.5 (3D), the Newtonian default; measured 2D/3D against LJ (`quadcopter/README.md`, F43–F55) | done |
| 13.6 | "Triangle of death" | 454–456 | Ground robots | new |
| 13.7–13.8 | Fixed-wing reachable space (13 m/s, 20° bank); single-attractor behaviours. Eq. 13.14 gives a ~101 m turn radius, which matches Fig. 13.16's loops; a coordinated turn would give ~47 m | 457–459 | `fixedwing/model.lua`, `ch13/UAV-Attractor.lua`: the three outcomes of Fig. 13.13 reproduced (F76) | done |
| 13.9 | Heading-particle torque law | 460–462 | `ch13/uav.lua` on the fixed-wing (Eqs. 13.15–13.18; repulsion turns away from the neighbour, F76) | done |
| 13.10 | Dumbbell agent | 462–463 | Pioneer | plan (corrected) |

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 13.11 | **Pioneer recipe: adaptive friction (13.23), caps, adaptive cap (13.24), static friction (13.25); 3–5-robot camera ring with three-point turns** | 463–465 | On the drones all measured: only η is needed; adaptive and static friction and the adaptive cap are not (F51, F54, F55). On the Pioneers (M2): adaptive and static friction do not settle the split law at D, and the cap has no collisions to prevent (F66, F68); ring on Pioneers to come | partly |
| 13.12 | 5 UAVs and 10 targets, gravity vs torque law | 465–467 | `ch13/UAV-Targets.lua`, 20 seeds: the torque law 1.6× the time over the targets and a tighter spread; not the lower turning (F76) | done |
| 13.13 | Four-step design procedure | 466–468 | Checklist | new |
| ★ | **Rotating formations for MAVs with a high minimum speed** (William, 2026-10-28): instead of each UAV circling alone, agents fly toward another agent, so the group rotates as one formation about a centre nobody chooses. Three versions on the fixed-wing model, against the book's lone loiter circles: (a) cyclic pursuit with IDs (i chases i+1; constant-speed vehicles settle into a regular rotating polygon with a fixed centroid, Marshall, Broucke and Francis 2004); (b) ID-free, each pursuing the nearest agent in a forward sector (the pursuit graph has one cycle per connected piece: rings, pair orbits, trailing followers; the sector width is the parameter); (c) Lennard-Jones with the torque law and nothing else, to see whether rotating "mills" form by themselves (D'Orsogna et al. 2006). The limit to state: equal speeds allow one ring, not a rigidly rotating lattice, and a ring of n at spacing D needs n D ≥ 2π R_min (≈ 13 UAVs at D 50 m, R_min 101 m); below that, lone circles win. Measures: spacing CV, closest approach, centre drift, settling time, and per mission: area covered and revisit time (surveillance, the capstone's job), every UAV keeping a neighbour in range (relay), staying in a region | — | `fixedwing/`, `ch13/uav.lua` | new |

## Ch. 14 — Adaptive learning, Daedalus (PDF 472–505) · `book/ch14.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 14.1 | Offline EA for three LJ laws (Table 14.2 target) | 475–484 | Particles; values from Fig. 15.1 | plan |
| 14.2 | 50 robots flow round a large obstacle | 483 | Particles → drones | partly |
| 14.3 | Offline law in the 5-goal corridor: 51.66% survive | 485–488 | Batch | plan |
| 14.4 | Line of sight between robots | 489–490 | G8 | plan |
| 14.5 | Mutation-rate groups, Approaches 1 and 2 (48.3 → 68.3%) | 491–497 | Daedalus loop | plan |
| 14.6 | Survival against obstacle density (Tables 14.7–14.10) | 497–500 | Batch, the cleanest numeric target | plan |
| 14.7 | Future: stuck robots switch to the gas model | 503 | ★ LJ → KT hybrid escape; compare with the chain-fountain escape (§7) | new |
| ★ | Chain-fountain escape: a free agent leads stuck agents out of a cul-de-sac as a chain | — | §7 | new |

## Ch. 15 — Statistical scouts (PDF 506–535) · `book/ch15.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 15.1–15.2 | Bernoulli and Bayes formulas (7 priors; closed form) | 511–517 | Library | plan |
| 15.3 | Navigation simulator: 800×600, 1,500 steps | 517–519 | The Ch. 14 world | plan |
| 15.4 | Ground truth from 1,000 runs, MSE | 519–524 | Batch (G3); environments with p = 0.1/0.5/0.9 to design | plan |
| 15.5–15.8 | Hypotheses 1–4 (prior, more scouts, scalability, difficulty) | 524–531 | Batch | plan |
| 15.9 | LJ scouts predict better than Newtonian scouts | 532 | Batch, both laws | new |
| 15.10 | Porting to real robots | 533 | ★ cascade crashes on drones | plan |

## Ch. 16 — Swarm design considerations (PDF 536–559) · `book/ch16.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 16.1 | Split-Newtonian law with a mass exponent | 540–541 | Force-law table (G9) | new |
| 16.2 | Two types make rings of any radius | 541–542 | Particles → drones; ★ spherical shells | new |
| 16.3 | Virtual particles | 542 | G9 | new |
| 16.4–16.5 | Asymmetric type-to-type laws; graph-based design | 542–550 | G9 | new |
| 16.6 | Resource protection (3 inner + 6 outer) | 543–552 | Demo; ★ drones guarding a VIP | new |
| 16.7 | Multi-field surveillance | 545–552 | Demo | new |
| 16.8 | Covert blind-spot tracking | 546–553 | Demo | new |
| 16.9 | Model-vs-vehicle metrics: suggested-vs-actual gap, settling time, lattice CV | 554–557 | **Acceptance metrics for the Ch. 13 work** | new |

## Ch. 17 — Swarm engineering (PDF 560–592) · `book/ch17.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 17.1–17.3 | Method; hexagon energy; single hexagon is the minimum | 560–572 | Theory and checks | new |
| 17.4 | Basic run: energy plateaus with defects | 572–574 | Particles | new |
| 17.5 | Method 1: synchronised drift | 575–577 | Particles | new |
| 17.6 | Method 2: edge-agent kicks, hexagon in every one of > 10,000 runs | 578–580 | Particles; ★ 3D; compare with Ch. 4.1 | new |
| 17.7–17.8 | Lossless flocking; the informed-agent bias limit (Eq. 17.24) | 580–585 | Sweep the goal-force share on existing formations | new |
| 17.9 | Quark model: coalescence 81%, informed chase, squeeze through a hole | 586–590 | Particles → velocity-commanded quads (G4) | plan |

## Ch. 18 — APO for global optimisation (PDF 596–620) · `book/ch18.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 18.1–18.2 | APO framework; APO1/2/3 laws | 600–606 | Headless Lua | new |

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 18.3 | 50-D benchmark, six functions (Table 18.2) | 606–609 | Batch, plus a control with the optimum shifted from the origin | new |
| 18.4 | Seven mass functions | 609–612 | Batch | new |
| 18.5–18.6 | Convergence theorem; choosing G | 613–616 | Closed-form checks | new |
| 18.7–18.8 | EAPO, VM-APO, LAPO; applications (no data) | 598, 616–618 | Notes | new |

### Ch. 19 — AP for noisy nonstationary environments (PDF 622–644) · `book/ch19.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| 19.1 | `apo.nlogo`: 7 particles find the Rastrigin minimum | 624–626 | Particles + G6 | plan |
| 19.2 | Noise 600,000 and 5,000,000 | 626–627 | Same | plan |
| 19.3 | N × D table (Tables 19.1–19.2) | 627–628 | **Main numeric target** | plan |
| 19.4 | Zoom: feature size against D | 627–629 | Same | new |
| 19.5 | Click to move the optimum (tracking) | 629–630 | ★ drones tracking a moving hotspot | plan |
| 19.6 | d + 1 minimum particles | 629–630 | 4 drones in 3D | new |
| 19.7–19.9 | APO vs PSO 2D: stationary, moving, rotated ellipse | 631–637 | Batch; PSO in Lua | new |
| 19.10–19.13 | 30-D trajectories; PSO → APO hybrid, hand-set and cyclic D | 637–643 | Headless | new |
| — | ★ Sensing from altitude (blur with height), 3D field | — | Our extension of 19.1–19.5 | plan |

### Appendix — Anomalous random() (PDF 646–652) · `book/ch20-appendix.md`

| # | Entry | Pages | BPP | Status |
|---|---|---|---|---|
| A.1 | Angle histogram with S extra calls (S = 25, 27 biased; ≥ 28 clean) | 646–648 | `checks/rng_appendix.lua` (G2): `math.random` biased at S = 3, 25, 26, 27; `lib/rng.lua` passes at every S | done |
| A.2 | Spatial histogram in the disk | 649–650 | Visual demo: math.random vs the new RNG | new |
| A.3 | Quarter-circle area (unaffected) | 651–652 | Control | new |
| A.4 | drand48 and others are fine | 651–652 | Choice of generator | new |

## 5. Ideas from the earlier chat, the plan and the thesis digests

These come from the conversation that produced `AP-3D-vehicles-plan.pdf`, from that plan, and from the thesis digests' sections on what they mean for the plan. Each one is missing from, or only touched on, in the tables above. They are grouped by the chapter they attach to. Source: **chat** = suggested in the earlier conversation; **plan** = a ★ item in the plan PDF; **digest** = a thesis digest (author named).

### Formations (Ch. 3, 4, 17)

| Idea | Source |
|---|---|
| ★ **Lag acts like heat:** measure how far drone lag shifts the G phase transition (Eq. 3.4), so the drone port gives a result as well as a demo | plan |
| ★ **Gentle annealing on drones:** random kicks that start large and cool to zero, to shake 3D lattices out of glassy arrangements. On drones the kicks are real accelerations, so keep them small | chat |

| Idea | Source |
|---|---|
| ★ **Choose N to suit the shape:** 13, 55 or 147 for icosahedra, or a count that completes whole hexagonal layers | chat |
| ★ **Three-label spin for FCC:** flight levels give ABAB (HCP) stacking; a three-label version of the spin could force ABC (FCC) | chat |
| ★ **Honeycomb's switching costs energy:** the square↔honeycomb switching that escapes the centre trap keeps drones moving. Measure the trade-off between avoiding the trap, energy and wobble | plan |
| ★ **Diamond caveats:** diamond is floppy under distance-only forces, so it is an open question whether the like-class links hold it rigid; the square-escaping switch needs rethinking, because the competing 3D arrangements differ | chat |
| ★ **Honeycomb at one altitude as ground coverage** for cameras or radio: the most area per drone for a given link distance | chat |
| ★ **Perfect Formation on drones:** the momentum checks show how much tracking lag disturbs the formation; test self-repair when switching formations; "One is Born" mid-flight becomes a launch problem; the shared compass fixes the cube's random rotation about the vertical | chat |
| ★ **Ch. 13 in 3D:** the chapter covers only wheeled robots and fixed-wing aircraft; quadcopters in 3D extend it, and its reachable-space analysis needs a second angle (climb rate) | chat |
| **Launch new drones from the side,** so a one-at-a-time launch doesn't climb up through the levels already formed | chat (open item) |

## Optimization and fluxotaxis (Ch. 8, 9, 19)

| Idea | Source |
|---|---|
| ★ **Missions for APO on drones:** finding a radiation hotspot or a gas source | chat |
| ★ **Readings from where the drone actually is,** not from its waypoint; downwash stirring a real plume as a noise source | chat |
| ★ **3D plumes:** buoyant gases rise and heavy ones hug the ground; puffs have height and reflect off the ground | chat |
| ★ **Wind changes with height** (slower near the ground); buildings have heights that make wakes and street canyons. Add a height-dependent mean wind as a 3D boundary condition with spin-up, and keep the obstacle mass-relocation rule against building walls | chat; digest (Zarzhitsky) |
| ★ **Source height:** a swarm spread over several altitudes finds the plume's vertical structure; a rooftop vent and a ground spill look the same in 2D. Report source height as a metric by repeating the main tables with the emitter raised | plan; digest (Zarzhitsky) |
| ★ **Wind from tilt and thrust** as a non-intrusive anemometer (fluxotaxis never ran on hardware for lack of one). Model its lag and noise; it may favour fluxotaxis or chemotaxis over anemotaxis | plan; digest (Zarzhitsky) |
| ★ **Downwash on the sensors:** bias each drone's sensed wind and concentration by its own rotor wash and compare with ideal sensors; test sensors mounted above and below the body (placement mattered on a single robot) | plan; digest (Zarzhitsky) |
| **Pseudo-source test:** buildings that create corner sinks or concentration pockets, where chemotaxis is fooled and fluxotaxis should not be | digest (Zarzhitsky) |
| **Flux stopping rule over a 3D shell** (sustained outflux for τ), which the thesis never enabled; the natural way to estimate source height | digest (Zarzhitsky) |
| **Re-derive G_t for 3D packing** (the thesis notes a 308 vs 231 discrepancy in its Eq. 4.7) | digest (Zarzhitsky) |
| **Compare directly with the app's 2D results:** success rate, time to find the source, distance flown; draw the plume as translucent puffs | chat |
| **Share the field code:** optimization and fluxotaxis both find a source in a field with a sensor term on top of the lattice force, so the 3D field sensing and drawing carry from one to the other | chat |

## Bioluminescence (Ch. 6)

| Idea | Source |
|---|---|
| ★ **Wake-triggered glow:** real bioluminescence is triggered mechanically, so the vehicles' own wakes light the water. Speed against stealth becomes a control problem | plan |

| Idea | Source |
|---|---|
| ★ **Dinoflagellates migrate vertically** by day and night, and currents add drift to everything. A 3D field of roughly 60 × 60 × 20 cells | plan; chat |
| ★ **Two underwater vehicles:** a multi-thruster ROV-style vehicle (like a BlueROV2) that behaves like an underwater quad, and a torpedo-shaped AUV that cannot hover or move sideways, the harder and more realistic case | chat |
| **Draw the glow** with translucent slices or point sprites | chat |
| **Underwater sensing:** sound at ~1500 m/s; the RF sync pulse won't propagate, so use an optical or clock-synchronised trigger | digest (Kunkel, Heil) |

## Coverage and kinetic theory (Ch. 4, 7, 11)

| Idea | Source |
|---|---|
| ★ **Volume coverage** (air sampling, searching a warehouse): in 3D the collision cross-section becomes an area, not a width | chat |
| ★ **Does the linear profile survive collisions that take time?** A quad needs half a second or more to change velocity, which should act like extra viscosity | plan |
| **Lag against collision time:** log commanded-vs-actual velocity lag (time to ~90%) relative to the mean time between collisions; the profile should hold only while the lag is much shorter. Sweep a controller gain to find the failure point | digest (Kerr) |
| **Track the backward-moving fraction** ($v_y < 0$), which Kerr blames for profile error, and the collision rate per bin | digest (Kerr) |
| **More 3D checks:** profile uniform in z; $\langle \tfrac{1}{2}mv^2 \rangle = (3/2)kT$; the full Maxwell speed histogram with a KS test (never tested); altitude bias from floor and ceiling | digest (Kerr) |
| ★ **Fixed-wing drones as gas molecules** (constant speed). Kerr's digest warns that equal speeds break the Maxwell speed distribution, so test the heading distribution instead | plan; digest (Kerr) |
| **Scale:** full quads manage ~100–200 in BPP against the app's 1,000 particles, so expect noisier histograms or use the point-mass vehicle (G5) | chat |

## Chains (Ch. 4.2, 12)

| Idea | Source |
|---|---|
| ★ **Where relays matter:** tunnels, mines, caves, parking garages, urban canyons; up a shaft or stairwell, over a building, down into a pit | chat |
| ★ **Chains for exploring maze-like spaces,** not only for holding a link | chat |
| ★ **"Does the relay hold?" as the score:** a link breaks when walls block the line of sight, not only on distance; the app's Max Distance gauge becomes a radio-range check | chat |
| ★ **Heading for a hovering drone:** the ellipse's axis comes from yaw or from a smoothed travel direction. It may decide whether chains stay straight or fold | plan |
| **3D chain details:** front tangential force in a random direction perpendicular to the heading; up/down IR sensors for shafts; acoustic line of sight blocked by walls while RF is not | digest (Maxim) |
| **Reproduce the 2D Maxelbot runs first:** 12 in/s, 3 IR sensors at 9″ range, 5 ft maximum spacing, the 14 ft and 8 + 12 ft corridors | digest (Maxim) |

## Learning and scouts (Ch. 14, 15)

| Idea | Source |
|---|---|
| ★ **Drones tune their own settings mid-mission** to cut losses on the obstacle field | chat |
| **Report "lost" (crashed) separately from "hit":** a crashed quad cannot be repaired at the goal. Repeat the safety-zone result (zero collisions, lower reachability) | digest (Hettiarachchi) |
| **Train easy, test hard:** the online world (obstacles ×3, > 16% coverage, Vmax × 1.5, 5 chained goals) is a ready difficulty knob for our field | digest (Hettiarachchi) |
| **Minimal Daedalus:** 13-float genome, worthiness 1000, rule copying on contact within 1.5R and in line of sight, 5% "panic" mutation when stuck, seed rates {1, 3, 5, 7, 9}%; feed the LJ output to the controller as a velocity or acceleration setpoint | digest (Hettiarachchi) |

| Idea | Source |
|---|---|
| **Hardware Hooke repulsion** (IR, range 30, front-arc sensors) as a template for realistic obstacle sensing | digest (Hettiarachchi) |
| **Scouts in two modes:** independent and in formation (same LJ as the swarm); test whether crash cascades break the thesis's "dependence helps" result | digest (Rebguns) |
| **Cost:** ground truth at ~1,000 runs per field is ~17 h with full drones; use G5 or fewer runs | digest (Rebguns) |

## Sensing (Ch. 10; G7)

| Idea | Source |
|---|---|
| **Three presets:** "design spec" (Heil, sub-inch within 6 ft), "Maxelbot" (0.6–0.7 in, 2.4–3.5 m, 8.33/N Hz) and "pessimistic" (Heil: σ 3.5 in, 14.5° bearing, drops past 0.6–0.9 m, cut-off at 6 ft) | digest (Heil, Maxim) |
| **Noise on each range, not on the position,** so geometric dilution at distance emerges by itself | digest (Kunkel) |
| **Couple sensing to attitude:** a tilting quad tilts its sound plane; receivers outside it drop out | digest (Heil) |
| **Speed-of-sound calibration parameter** (matters most underwater) | digest (Heil) |
| **Relaying out-of-range positions** with a hop-count "doubt level" (Heil App. B) | digest (Heil) |
| **Cheap baseline:** sector-only bearing (±30°) against full trilateration | digest (Heil) |
| **The 50-sample running average adds lag** that matters for kinetic theory and Daedalus | digest (Heil) |
| **Drops:** 1% acoustic and 0.2% RF; multiplicative U[0.9, 1.1] noise | digest (Maxim) |
| **3D receiver:** a fourth at (0, 0, d), giving $z = (b^2 - e^2 + d^2)/2d$ (our extension) | digest (Heil, Maxim) |

## Sound, BPP and the app

| Idea | Source |
|---|---|
| **Exaggerated pitch:** stretch pitch deviations ~3× on the `emphasis` slider so a climb or hard turn becomes an audible swoop | chat (not built) |
| **Event sounds:** a knock and motor wind-down on a crash; a short chirp on launch and landing | chat (not built) |
| **Info dock pane** in BPP instead of the status line | chat (open item) |
| **Port the Uniform Coverage fixes to the Android app** (±15° cone; dead-end retry) | plan |
| **FIXES item 27** (precompiled-header build bug): documented, not fixed | chat (open item) |

The plan's capstone mission is now Section 6, extended to ground, surface and underwater vehicles.

## 6. The capstone: one mission, five kinds of vehicle

Ch. 13's central point is that physicomimetics only decides where each agent should go, and each vehicle's own controller works out how to get there. The book's experiments span several vehicle types (Maxelbots and Pioneers on the ground, surface boats, fixed-wing aircraft in simulation), but never all of them in one study. Running the same force laws on five kinds of vehicle, in one mission, makes the claim that the method is not vehicle-specific directly.

The plan's original capstone was drones only: launch in thirds, a perfect 3D lattice by compass and address, scouts probing a city, fluxotaxis finding a plume's source and height, and a chain relay back to the launch point. The extended mission keeps all of that and adds the other vehicles.

### The mission: a harbour incident

A leak from a ship releases a chemical into both the air and the water.

| Vehicle | Its part in the mission | Book chapters |
|---|---|---|
| Quadcopters | Launch in thirds, form a perfect 3D lattice, trace the airborne plume to its source and height | 4.1, 8–9 |
| Fixed-wing aircraft | Circle over targets for wide-area surveillance, since they cannot hover | 13 |
| Ground robots | Scouts test the dock area first; a chain relay goes into a warehouse or storm drain; robots stuck in the drain are led out as a chain (§7); a kinetic-theory sweep covers the sheds | 15, 12, 7, ★§7 |
| Surface boats | Map the slick or the bioluminescence; find the darkest route for a covert approach | 6 |
| Underwater vehicles | Trace the plume underwater; hold a relay chain from the surface down to depth | 8–9, 12 |

Across media, the surface boats relay between sound underwater and radio above it. Ch. 16's typed force laws let different vehicle types share one formation, and depth bands work for underwater vehicles the way flight levels work for the drones.

### New tasks

| Task | Needs | Effort |
|---|---|---|
| ★ **Cross-vehicle benchmark** (started, `BENCHMARK.md`: quadcopters, ground robots, boats and the ideal particles, tasks 1–2; F68): the same task with the same force-law settings on all five kinds of vehicle (quadcopter, fixed-wing, ground robot, surface boat, underwater vehicle), reporting the Ch. 16.9 measures for each: settling time, lattice quality, losses, and the gap between suggested and actual position | G18, G10, G11, G12 | medium |
| ★ Benchmark task 1: formation (2D lattice; 3D for the vehicles that can) (done for four vehicles) | G18 | easy per vehicle |
| ★ Benchmark task 2: goal through obstacles (Ch. 3.6) (done for four vehicles) | G18 | easy per vehicle |
| ★ Benchmark task 3: uniform coverage (Ch. 11) | G18, G4 | medium |
| ★ Benchmark task 4: chain formation (Ch. 12) | G18, G8 | medium |
| ★ ROV-style underwater vehicle (holonomic, hovers) | G11, G19 | medium |
| ★ Torpedo-shaped AUV (cannot hover or move sideways; heading particle) | G11, G19, G18 | medium–hard |
| ★ Underwater formation in depth bands (the counterpart of flight levels) | G11, G19, G20 | medium |
| ★ Underwater plume tracing, the plume carried by a current | G14, G19 | hard |
| ★ Relay chain from the surface down to depth, over sound | G20, Ch. 12 | medium |
| ★ Mixed-vehicle formations with typed force laws (Ch. 16) | G9, G18 | medium |

| Task | Needs | Effort |
|---|---|---|
| ★ Surface boats as sound-to-radio gateways linking underwater and air swarms | G20, G15 | medium |
| ★ Harbour features of the world generator, and the full mission script | G21, everything above | hard |

The benchmark matrix is four tasks × five kinds of vehicle, i.e. 20 runs, each over several seeds. That table is the scientific result; the mission demonstrates it.

## The world generator (G21)

Our own generator, designed around this mission and serving every chapter. It is 3D and modelled on how real places are built, in general rather than any particular place. **Nothing is taken from the old map makers** in the CPT code (`MakeMap.c` for Textron, `MapGen.c`); `lib/world.lua` only *writes* their file format (`writeCPT`), so Zarzhitsky's PlumeGen and TraceGen can run on our worlds as an independent check of our plume port.

**Approach (decided 2026-10-17): procedural, calibrated to published statistics of real places.** Importing real places (OpenStreetMap footprints with `building:levels`, elevation models, port depth charts) is future work; it would then also calibrate the statistics.

**Layers, in build order:**

| Layer | What | Based on |
|---|---|---|
| Terrain (**done**, `lib/terrain.lua`) | Heightfield: coastal plain, hills on it fading toward the shore, river valleys; one rule keeps every dry slope drivable (≤ 15% along the grid) and also shapes valley walls and banks | Fractal value noise; meanders by the disturbed periodic model (Ferguson 1975, 1976), with neck cutoffs: sinuosity 1.4, wavelength 11 widths (Leopold and Wolman 1960), irregular, as natural meanders are (F64) |
| Water (**done**, `lib/terrain.lua`) | Sea, harbour, rivers, ponds with depth; dredged approach, channels, turning basin, berths with 1:3 sides, quay walls vertical; shelf shoaling to the shore | Los Angeles main channel 53 ft (16.2 m); Long Beach approach 76 ft (23.2 m): ours 24 m and 16 m |
| Zones (**done**, `lib/town.lua`) | Districts around centres, density falling exponentially outward, typed by Local Climate Zone (compact/open high-, mid-, low-rise, large low-rise, sparsely built, heavy industry; trees, fields, paved, soil, water); districts for ports, parks, industry; woods a stated share of open land | Clark (1951); Stewart and Oke (2012), building fraction: compact high-rise 40–60%, > 25 m; open low-rise 20–40%, 3–10 m; sparsely built 10–20% (corrected, F61) |
| Streets (**done**, `lib/town.lua`) | Arterials through everything, grids in cores, curving collectors and cul-de-sacs in suburbs, winding rural roads; bridges; every street connected to the arterials | Manhattan 1811: streets 60 ft every ~264 ft, avenues 100 ft every ~900 ft (80 × 274 m); Chicago 66 ft streets, 100 × 200 m; fire code: dead ends to 750 ft, 96 ft turnaround |
| Lots, buildings (**done**, `lib/buildings.lua`) | Frontage lots with setbacks; rectangles and L, T, U footprints along streets; whole floors (3.5 m), heavy-tailed, lower toward the edge; farmsteads; the gap rule; each zone's building fraction and mean height inside the published LCZ ranges | Schläpfer et al. (2015): one-storey buildings peak at 3.5 m, heavy tail, height falls from the centre; US median new lot 8,400 sq ft (2023); Stewart and Oke (2012) ranges (F62) |
| Vegetation (**done**, `lib/vegetation.lua`) | Woods of trees over zone A (300 a hectare, closed above and open below the crowns, so drones fly between the trunks; F70), hedgerows on field boundaries, street trees, garden and park trees (trunk and raised crown); trees block sight and sound, are porous to air (for the plume, later) | LCZ land covers; canopy 20 m, mature woodland 200-400 trunks a hectare, street trees 12 m apart, fields 250 m |
| Mission features (**done**, `lib/features.lua`) | Quay, piers, ship at a berth (the leak), warehouses with doorways, container yard with pockets, storm drain and outfall, launch pads | The capstone; 20 ft containers (6.1 × 2.44 × 2.59 m) |

**Presets** (parameter tables for one generator): **rural** (fields, woods, hedgerows, farmsteads, a river and pond, winding roads, rolling terrain); **suburban** (houses on lots, curving streets, cul-de-sacs, trees, a park with a pond);

**city centre** (compact mid- and high-rise grid, heights falling from the centre, plazas, a park); **industrial port** (warehouses, container yard with pockets, quays, dredged basins); **harbour town**, the capstone (all of these around a harbour with real depth structure, the ship at its berth, storm drains and outfall, launch pads).

**Everywhere:** gaps between buildings fit an agent or are closed (the Android port's rule); worlds are seeded from their own stream, so a seed always gives the same world and sweeps vary one knob at a time; distant districts are merged into static meshes, with full detail only in the mission area.

**Outputs (done):** BPP geometry (terrain, water and seabed, buildings, trees, the drain's manholes); grids of ground height, water depth, zone and occupancy at a chosen height (`raster`, `writeRaster`; `writeCPT` for Zarzhitsky's tools); line of sight (`sight`); metric masks (cells, shadow cells, `traps`, water). **Later:** wind with height and currents.

**The book's worlds stay as the book draws them,** as presets for the chapter ports: Ch. 3's 12 discs, Ch. 4's coverage box, Ch. 7's corridor (Fig. 7.2's shapes), later Ch. 11's environments, Ch. 12's maze and Ch. 14's field. The core and these presets came in M0 (`lib/world.lua`, `Worlds.lua`); the realistic layers follow in the table's order, one patch each: terrain and water are done (`lib/terrain.lua`; the five presets in `lib/places.lua`, shown by `Worlds.lua` presets 5–9; `checks/terrain.lua`), zones and streets (`lib/town.lua`, `checks/town.lua`), lots and buildings (`lib/buildings.lua`, `checks/buildings.lua`), and the mission features and vegetation (`lib/features.lua`, `lib/vegetation.lua`, `checks/features.lua`). All five presets are complete (FINDINGS F60–F63).

### Order

Build G18 and the cross-vehicle benchmark early, right after the Ch. 13 fixes, starting with ground robots and surface boats because they are simpler than underwater vehicles. That tests the claim cheaply. The full mission comes last, since it needs nearly everything else.

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## 7. Escaping cul-de-sacs: the chain fountain

**The gap in the book.** Lennard-Jones swarms flow round obstacles, but agents caught in a concave pocket stay stuck. Ch. 14 leaves this as future work and suggests that stuck robots switch to the kinetic-theory gas model to escape (§14.11, PDF 503). Hettiarachchi's thesis also lists cul-de-sacs as an open problem and never defines "stuck".

**The idea.** `demo/WyomingWill/ChainFountain/chain-fountain.lua` simulates a bead chain pouring out of a jar (the Mould effect). The part of the chain already outside pulls the rest over the rim; links are picked up off the pile one at a time, and because a joint can only bend so far (30° in the exhibit), the chain arches up and over the rim instead of dragging across it. The same mechanism can lead stuck agents out of a cul-de-sac, as a chain behind an agent that is not stuck. This is an alternative to the book's chain formations (Ch. 4.2, 12), which deploy a relay outward from a base; this chain pulls agents out of a trap.

| Chain fountain | Escaping a cul-de-sac |
|---|---|
| The jar | The cul-de-sac |
| The end hanging outside | A free agent (not stuck) as the head |
| Gravity on the falling chain | The goal force, felt only by agents already outside |
| Links picked up one at a time | Stuck agents join the chain's tail in order, nearest first |
| Fixed link length; joints bend at most 30° | Each agent holds a set distance behind the one ahead, with a limit on the angle between successive links |
| The arch over the rim | Clearance at the trap's mouth |

**Why it might work well:**

- **The pull grows as agents get out.** In the fountain the weight of the falling chain drives the flow, so the more is outside, the stronger the pull. If only agents outside the trap feel the goal force, the swarm behaves the same way.
- **The bend limit stops corner-cutting.** Plain follow-the-leader cuts corners and drags followers into walls; a bend limit makes the chain swing wide round the mouth.
- **It suits real vehicles.** A bend limit is in effect a minimum turn radius, which ground robots and fixed-wing aircraft have anyway (Ch. 13). In 3D, drones can do what the bead chain does: arch up and over the wall rather than thread back out through the mouth.

**What the fountain's latest runs add.** The arrangement of the pile decides the arch. Poured as a jumbled heap, as in the experiments, a ball chain stands at $h_2/h_1$ = 0.11–0.15 in Bullet against 0.12–0.14 measured (Biggins and Warner 2014; Pantaleone 2019). Laid as the exhibit's tidy coil, the same chain is picked up at the wall, rides the rim and stands at only 0.06 (commits `1aa427a`, `f938b01`). For the swarm this suggests two things:

- **The order of pick-up matters.** If stuck agents lie along the trap's wall and the nearest one always joins the tail, the chain will ride the wall, like the coil riding the rim. Recruiting from inside the cluster, away from the walls, may give the clearance the heap gives.
- **Only the agents near the pick-up need to act.** The fountain now simulates only the links within 4 cm of the pick-up and keeps the rest of the heap fixed, and still matches the full heap. In the same way, only stuck agents near the chain's tail need to take part, so the method stays local as the swarm grows.

**What doesn't carry over.** The fountain's push comes from the pile pushing back on each link as it is lifted. Agents have no pile, so the arch height will not follow Biggins and Warner's relation for $h_2$. Whether a virtual version of that push helps is an experiment, not a given.

## Tasks

| Task | Needs | Effort |
|---|---|---|
| ★ **Cul-de-sac world:** U-shaped and deeper pocket obstacles added to the Ch. 14 obstacle field, and a definition of "stuck" (no progress toward the goal over a time window) | G13 | easy |
| ★ **Virtual chain law:** set distance to the agent ahead, bend limit, recruitment of the nearest stuck agent at the tail, goal force only outside the trap. 2D particles first | G9 | medium |
| ★ **Compare escapes:** plain Lennard-Jones (baseline), kinetic-theory escape (Ch. 14.7), the chain-fountain escape, and the Ch. 12 chain law. Measures: share reaching the goal, time, collisions, agents out per second, clearance at the mouth | G3, G16 | medium |
| ★ **Sweeps:** link length, bend limit and head speed; whether the outflow rate settles to a steady value as the fountain's does | G3 | easy |
| ★ **Pick-up order:** stuck agents strung along the wall against clustered inside the pocket, and recruiting the nearest agent against one away from the walls; measure clearance at the mouth and wall contacts (the coil-versus-heap result) | G3, G8 | easy |
| ★ **Vehicles:** ground robots round corners; quadcopters arching over the wall in 3D; fixed-wing aircraft, whose turn radius is the bend limit | G10, G12, G18 | medium |
| ★ **In the capstone:** ground robots stuck in the docks or the storm drain, led out by a free one (§6) | §6 | — |

## 8. Build order

This revises the plan's order using what the book says.

Before porting a task, read its chapter note, read its `.nlogo` in `netlogo/`, examine the Android app's version (mostly built from the `.nlogo`; sometimes improved, sometimes wrong), and read the matching thesis digest. The book's chapters leave out values that the models and theses contain. `WORKFLOW.md` describes the steps every task goes through.

| Step | Work | Why now |
|---|---|---|
| 0 | **G2 shared RNG** (move the existing multiply-with-carry generator out of `Uniform_Coverage.lua`, add a seed setting) **+ G3 batch runner + G1 monitors + G21 world generator** (core and the first book presets), then Ch. 2's three tests | Every measurement and every world after this depends on them |
| 1 | **Ch. 13 on the drones:** η → adaptive friction → static friction → adaptive cap, each switchable, measured with the Ch. 16.9 metrics on Newtonian 2D/3D over several seeds | The plan's step 1, now with four parts and acceptance metrics |
| 2 | ★ **G18 vehicle interface + cross-vehicle benchmark**, first with ground robots (G10) and surface boats (G11) alongside the quadcopters: formation and goal through obstacles (§6). **Done (M2):** the claim holds for the holonomic vehicles; ground robots fail only through the minimum wheel speed (`BENCHMARK.md`, F68) | Tests the vehicle-independence claim early and cheaply |
| 3 | Ch. 3 fidelity checks (G_t sweep incl. 3D, Disable One, Hooke, honeycomb) and Ch. 4.3/Ch. 11 coverage against Tables 4.2, 11.1 and Fig. 4.19 | Cheap; firms up what is already "done" |
| 4 | Ch. 4.1 perfect square → cube, then the triangle; Ch. 17 Method 2 as the alternative | The plan's step 2 |
| 5 | G4 velocity commands + G5 point mass; Ch. 4.4 Couette and Ch. 7 theory checks | Opens all the kinetic-theory and quark work |
| 6 | Ch. 14 offline world → Ch. 15 scouts (particles), then Daedalus; add cul-de-sacs and compare escapes: kinetic theory (Ch. 14.7) against the ★ chain fountain (§7) | The plan's step 3, in the order the book's dependencies need; the cul-de-sac is the book's open problem |
| 7 | Ch. 19 formation APO with G6 fields → ★ from altitude, moving hotspot | The plan's step 4 |
| 8 | Ch. 5 LOPF and Ch. 16 type table: square, line, star, V, rings, protection | New, cheap on the formation code |
| 9 | G7/G8/G10 sensing and ground vehicles → Ch. 10 checks, Ch. 11 and 12 replicas, Ch. 12 chains, then ★ 3D shafts | Hardware-faithful track |
| 10 | Ch. 8 unit tests → G14 plume → Ch. 9 → ★ fluxotaxis 3D | Flagship |
| 11 | G11 ASV → Ch. 6 → ★ ROV and AUV (G19, G20); G12 fixed-wing → Ch. 13.7–13.12; complete the cross-vehicle benchmark with all five kinds of vehicle; Ch. 1 demos; Ch. 18 benchmark | Remaining vehicles and chapters |
| 12 | ★ **Capstone:** the world generator's harbour (G21), underwater plume and relay, mixed-vehicle formations, the full mission (§6) | Needs nearly everything else |